

Paper 29.25 - Monster Horizon Toolkit Supplement

CQE-paper-29.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-29.25.md

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Purpose

This supplement shows how to reproduce Paper 29's Monster/energy-bound quarantine. The proof is in Paper 29 and its formal verifier. The toolkit is a reader aid for checking which rows are closed and which rows remain horizon hypotheses.

Digital Route

Run:

```
`python production/formal-papers/CQE-paper-29/verify_monster_energy_bound_hypotheses.py`
```

The expected status is ``pass_with_quarantined_hypotheses``. The verifier checks ``47*59*71 = 196883``, checks ``196884 = 1 + 196883``, verifies the finite VOA partition $Z(q) = 2q^0 + 6q^5$, and confirms that all physical energy-bound readings remain unwitnessed.

Analog Route

Place three factor cards labeled ``47``, ``59``, and ``71``. Multiply them into a single binding card labeled ``196883``. Add a separate observer bead labeled ``1``, then write ``196884`` on the combined McKay card.

Lay out eight chart-state cards. Mark ``(0,0,0)`` and ``(1,1,1)`` as weight 0. Mark the other six cards as weight 5. The white-thread result is $Z(q) = 2q^0 + 6q^5$.

Now place black tags on three attempted promotions: "weight 5 is physical energy", "196883 is a universal ceiling", and "Pariah closure is a physical law". Leave those strings loose until a witness function exists.

Hidden-Guess Diagnostic

Training mode should hide the label until after the reviewer chooses one:

- closed arithmetic row,
- closed finite VOA partition row,
- valid energy analog,
- invalid physical-energy promotion,
- invalid universal-ceiling promotion,
- invalid physical-boundary promotion,
- open horizon obligation.

The answer key distinguishes replayable finite evidence from unwitnessed physical interpretation.